

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)
DepartmentofArtificialIntelligenceandDataScience ASSESSMENTTOOL2
–SCENARIOBASED

CourseCode:	DSA03	CourseTitle:	NaturalLanguageProcessing
COAssessed:	CO2 – Manipulation	Assessment:	AssessmentTool2–SCENARIOBASED
Weightage:	30% of		
CO5Statement:	Designinflectional,derivational,finite-stateparsing,stemmingalgorithmsandforanalysemorphological methods. (BL-3)		
Student Name:	Y. Harsha Vardhan	Reg. No.:	192425227
Section/Batch:	B.TechAI-ML	Date:	30-07-2026

CodeofConduct

I Y. Harsha Vardhan Reddy (Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

EnglishWordClasses, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
EnglishWordClasses, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms and semantic effects of derivation.	Q2
Counting Words, EnglishWordClasses, Part of Speech Tagging	Morphology-based normalization of govern, government governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, EnglishWordClasses, Part of Speech	Inflectional normalization of create, creates, and creating; extraction of root forms and classification Tagging tense/aspect transformations.	Q5

1. MorphologicalProcessingforSearchEngine

InputWords

■ Analyzing

■ Analysis

■ Analytical

Step1:MorphologicalBreakdown

Word		
Analyzin g		
Analysi s		
Analytica l	Analyz e	

-ing

-sis

-ical

Step2:Classification

■ Analyzing→Inflectional(Presentparticiple)

- Analysis→Derivational(Formsanoun)
- Analytical→Derivational(Forms anadjective)

Step3:Normalization

Allwordsaremapped tothecommonbaseform:

Analyze

FinalOutput

- Analyzing=Analyze+ing
- Analysis=Analyze+sis
- Analytical=Analyze+ical

NormalizedRoot=Analyze

2. MorphologicalParsingforSentimentAnalysis

Step2:Classification

- Disagree→Derivational
- Agreement→Derivational
- Agreeable→Derivational

InputWords

- Disagree
- Agreement
- Agreeable

Meaning

- dis-givesoppositemeaning.
- -mentconvertsverbintonoun.
- -ableconvertsverbintoanadjective.

Step1:MorphologicalBreakdown

Word		NormalizedRoot
Disagree		Agree
Agreement		
Agreeable		

— agree -ment

agree -able

3. Morphology-Based Normalization

Input Words

■ Govern

■ Government

■ Governance

Step 1: Morphological Breakdown

Word	■ Government → Derivational ■ Governance → Derivational
Govern	
Government	Normalization Gov rAll words are converted into the common base form:
Governance	Govern

Final Output

■ Govern = Govern

■ Government = Govern + ment ■ Governance =

Govern + ance Normalized Root = Govern

Step 2: Classification

■ Govern → Base word

— -ment

-ance

4. Morphological Parsing for Document Classification

Normalization

All words are mapped to:

Act

Input Words

- Activate
- Activation
- Reactivation

Final Output

- Activate = Act + ivate
- Activation = Act + ivation
- Reactivation = Re + Act + ivation

Step 1: Morphological Breakdown

Word	
Activate	
Activation	
Reactivation	

Normalized Root = Act

Step 2: Classification

- Activate → Derivational
- Activation → Derivational
- Reactivation → Derivational

Act -ivate

Act -ivation

Meaning

- re- means "again".
- -ation changes verb into noun.

5. Morphology-Based Normalization

Input Words

- Create
- Creates
- Creating

Step 1: Morphological Breakdown

Word	All words are mapped to: Create
Create	Final Output
Creates	■ Create=Create ■ Creates=Create+s
Creating	■ Creating=Create+ing NormalizedRoot=Create

Step2:Classification

- Create→Baseword
- Creates→Inflectional(Presenttense) ■
- Creating→Inflectional(Presentparticiple)

Normalization

Conclusion

- -s
- ing

Morphological analysis helps identify roots, prefixes, and suffixes while distinguishing between inflectional and derivational forms.

Rubrics for Calculation

Criteria	Weightage	(Example)				Clarity Response	15%	Response structure and logic
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario	Application of Concepts	25%	Applies concepts effectively to address the problem			
			Decision Making	20%	Makes reasoned decisions based on strong evidence			
Identification of Key Issues	20%	Accurately identifies relevant issues involved in the scenario						

scenario

Q.No./ Criteria Level	Understandin gof Scenario	Identification of KeyIssues	Applicati onofCon cepts	Decisio n Making	Clarity Respons e	ofSut	Total %
1							
2							
3							
4							
5							

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature:	Signature:	Signature:
Date:	Date:	Date: